Lab Worksheets – 2307421L- Ubiquitous Sensing, Computing and Communication (USCC)

Practical NO. 1: Sensor Data Logging & Serial Communication

Batch: Batch 1

Group No.: 4

Members:

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Objective:

- To interface multiple agricultural sensors (DHT11, Soil Moisture, and pH Sensor) with the ESP8266 microcontroller.
- To capture real-time data for temperature, humidity, soil moisture, and soil pH levels.
- To display the collected sensor data on the Serial Monitor using serial communication.
- To analyze and understand environmental parameters affecting plant growth.
- To prepare the system for integration with cloud platforms for IoT-based monitoring in future applications.

Tools & Technology Used:

Tool / Component	**Purpose**
ESP8266	Microcontroller for data sensing and Wi-Fi communication
DHT11 Sensor	Measures temperature and humidity
Soil Moisture Sensor	Measures soil water content
pH Sensor	Monitors soil acidity/alkalinity

Arduino IDE	Coding and uploading firmware
Firebase	Cloud-based data logging
USB Cable & Jumper Wires	Hardware connections

Procedure:

1) Connect DHT11, soil moisture, and pH sensors to ESP8266 GPIO pins.

2) Configure ESP8266 to connect to Wi-Fi.

3) Initialize Firebase connection using API key and database URL.

4) Continuously read sensor data using analog and digital pins.

5) Display readings on the Serial Monitor.

6) Send real-time sensor values (Temperature, Humidity, Moisture, pH) to Firebase.

7) Verify that data is updating correctly in Firebase.

Code Snippets / Configuration:

```
#include <ESP8266WiFi.h>

#include <Firebase_ESP_Client.h>

#include <DHT.h>

// ----------------- WiFi & Firebase Credentials -----------------

#define WIFI_SSID "Airtel_Mighty Raju"

#define WIFI_PASSWORD "Mitaoe#701"

#define API_KEY "AIzaSyAUxLwgUALpPcWpp7_5GwHyNHsgNJQO3O8" // your full key

#define DATABASE_URL "https://smart-agriculture-df146-default-rtdb.firebaseio.com/"

// ----------------- Sensor Pins -----------------

#define DHTPIN D4

#define DHTTYPE DHT11

#define ANALOG_PIN A0
```

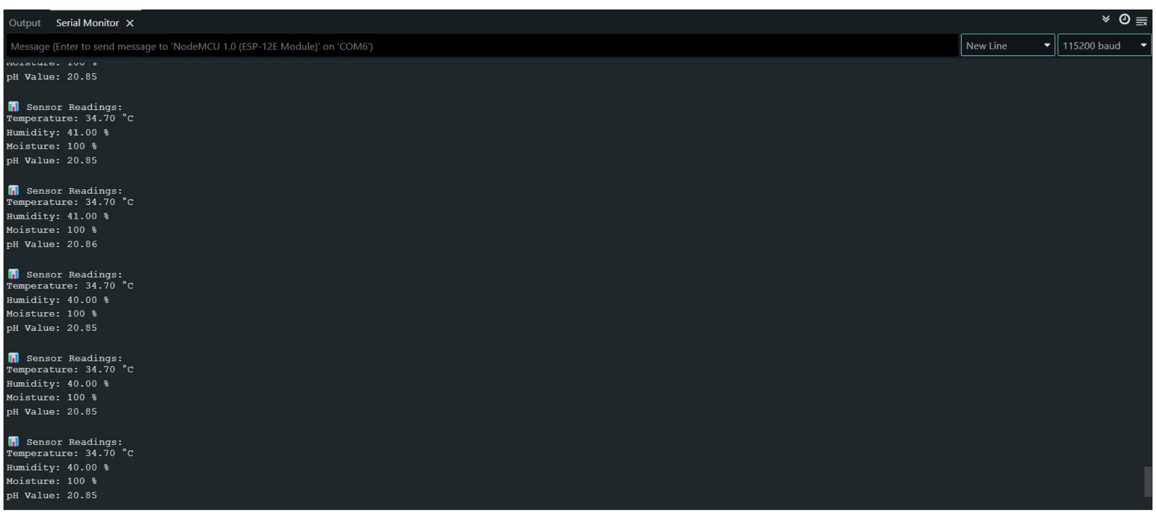
```
DHT dht(DHTPIN, DHTTYPE);

FirebaseData fbdo;

FirebaseAuth auth;

FirebaseConfig config;

// ----------------- Setup -----------------

void setup() {

  Serial.begin(115200);

  delay(1000);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  Serial.print("Connecting to WiFi");

  while (WiFi.status() != WL_CONNECTED) {

  Serial.print(".");

  delay(500);

  }

  Serial.println("\n✅ WiFi connected!");

  // Firebase configuration

  config.api_key = API_KEY;

  config.database_url = DATABASE_URL;

  // Anonymous sign-in

  if (Firebase.signUp(&config, &auth, "", "")) {

  Serial.println("✅ Firebase sign-up successful!");

  } else {

  Serial.printf("❌ Sign-up Error: %s\n", config.signer.signupError.message.c_str());

  }

  Firebase.begin(&config, &auth);

  Firebase.reconnectWiFi(true);

  dht.begin();

  Serial.println("✅ Firebase initialized!");
```

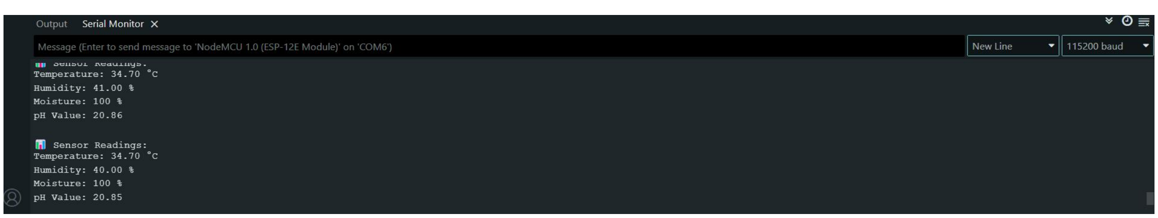
```
}

// ----------------- Sensor Functions -----------------

float readPH() {

  int samples = 10;

  float total = 0;

  for (int i = 0; i < samples; i++) {

  total += analogRead(ANALOG_PIN);

  delay(10);

  }

  float avgADC = total / samples;

  float voltage = avgADC * (3.3 / 1023.0);

  float phValue = 7 + ((2.5 - voltage) / 0.18);

  return phValue;

}

int readMoisture() {

  int val = analogRead(ANALOG_PIN);

  return map(val, 1023, 0, 0, 100);

}

// ----------------- Main Loop -----------------

void loop() {

  float temperature = dht.readTemperature();

  float humidity = dht.readHumidity();

  int moisture = readMoisture();

  float pH = readPH();

  Serial.println("\n📊 Sensor Readings:");

  Serial.printf("Temperature: %.2f °C\n", temperature);

  Serial.printf("Humidity: %.2f %%\n", humidity);

  Serial.printf("Moisture: %d %%\n", moisture);

  Serial.printf("pH Value: %.2f\n", pH);
```

```
  if (Firebase.ready()) {

  // Create unique path using timestamp

  String path = "/SmartFarm/Readings/" + String(millis());

  Firebase.RTDB.setFloat(&fbdo, path + "/Temperature", temperature);

  Firebase.RTDB.setFloat(&fbdo, path + "/Humidity", humidity);

  Firebase.RTDB.setInt(&fbdo, path + "/Moisture", moisture);

  Firebase.RTDB.setFloat(&fbdo, path + "/pH", pH);

  Serial.println("✅ Data uploaded to Firebase!");

  }

  delay(5000);

}
```

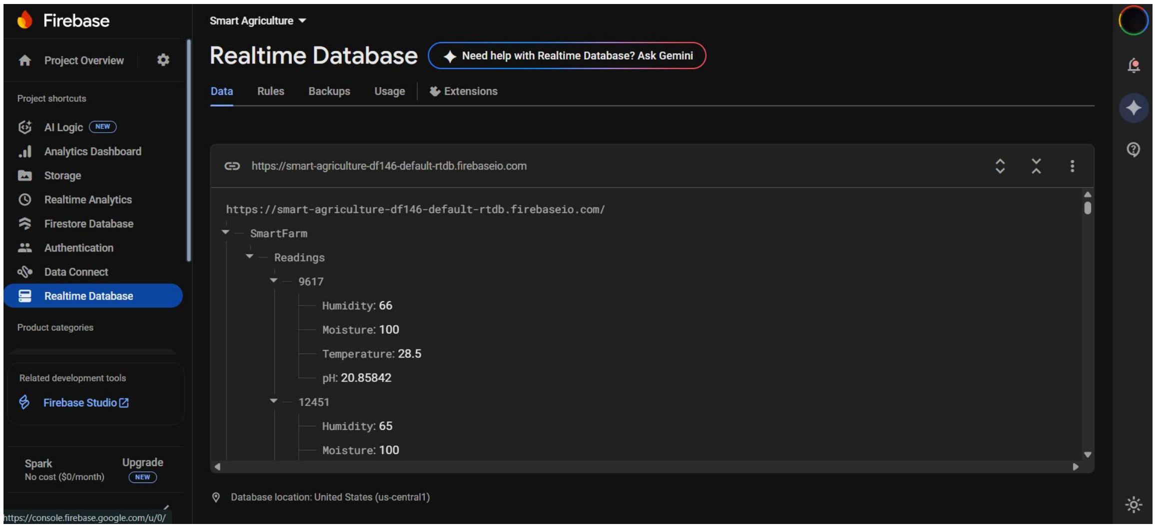
Observations / Result:

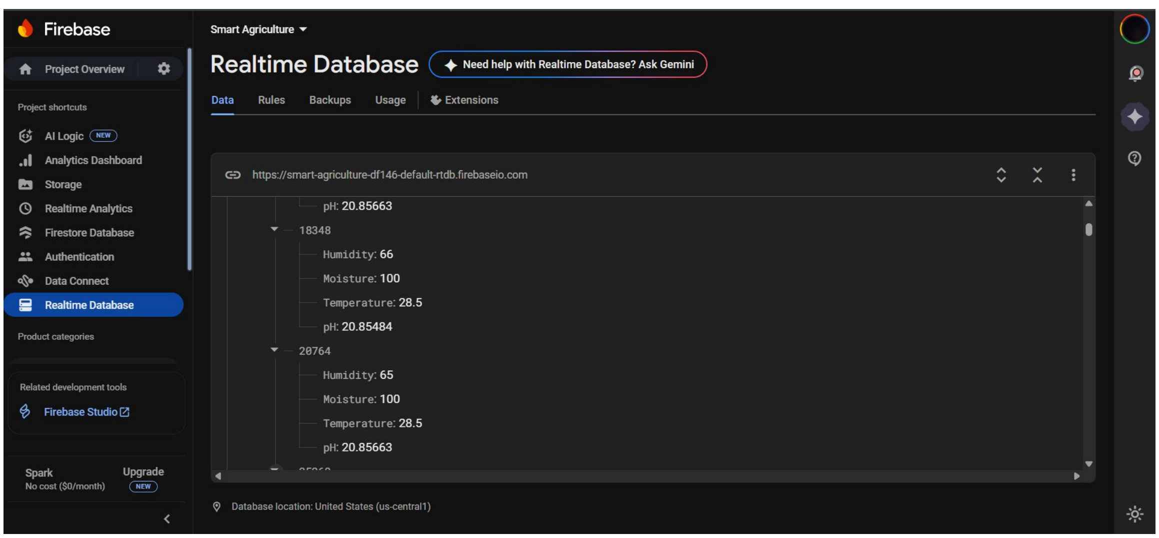
1)Sensor readings displayed in the Serial Monitor:-





2) Data updated in real time in Firebase Realtime Database:-





Troubleshooting & Debugging:

· **Issue:** DHT11 showing “nan” values.
Fix: Checked sensor connections and added delay after initialization.

· **Issue:** Moisture sensor giving unstable readings.
Fix: Used analogRead averaging technique.

· **Issue:** pH sensor giving noisy output.
Fix: Calibrated with standard buffer solution and added filtering.

Conclusion:

The Smart Agriculture System successfully monitors environmental and soil parameters essential for crop growth. This practical helped in learning sensor interfacing, serial data communication, and real-time monitoring, which can be extended for IoT-based smart farming applications.